

Node.js Important Questions & Detailed Notes

1. What is Node.js?

- Node.js is an open-source and cross-platform JavaScript runtime environment.
- It allows JavaScript code to run outside the browser.
- It is built on the V8 JavaScript Engine developed by Google.
- It is mainly used for backend development, APIs, and real-time applications.

2. What are the features of Node.js?

- Asynchronous and event-driven architecture.
- Single-threaded but highly scalable.
- Fast execution because of the V8 Engine.
- Non-blocking I/O operations.
- Large package ecosystem through npm.

3. Explain the Event Loop in Node.js.

- The Event Loop is responsible for handling asynchronous operations.
- It checks whether the call stack is empty and executes pending callbacks.
- It allows Node.js to handle many requests simultaneously without creating multiple threads.

4. Difference between Blocking and Non-Blocking I/O.

- Blocking I/O waits until an operation finishes before executing the next statement.
- Non-Blocking I/O continues executing other operations while waiting for the result.
- Node.js primarily uses non-blocking operations for better performance.

5. What is npm?

- npm stands for Node Package Manager.
- It is used to install and manage third-party packages.
- The package.json file stores dependencies and project information.

6. What is a Module in Node.js?

- A module is a reusable piece of code.
- Node.js provides built-in modules such as fs, http, path, and os.
- Developers can create custom modules and install external modules.

7. Explain the fs module.

- The fs module is used to work with files and directories.
- It can create, read, update, and delete files.
- It supports both synchronous and asynchronous methods.

8. What is Express.js?

- Express.js is a web application framework built on Node.js.
- It simplifies routing, middleware, and API development.
- It is widely used to build REST APIs and full-stack applications.

9. What is Middleware?

- Middleware functions execute during the request-response cycle.
- They can modify requests and responses.
- Middleware is commonly used for authentication, logging, and error handling.

10. What is REST API?

- REST stands for Representational State Transfer.
- It enables communication between client and server using HTTP methods.
- Common methods are GET, POST, PUT, PATCH, and DELETE.

11. What is JWT?

- JWT stands for JSON Web Token.
- It is used for secure authentication and authorization.
- A JWT contains Header, Payload, and Signature.

12. What are Streams in Node.js?

- Streams process data in small chunks instead of loading the entire file into memory.
- Types of streams are Readable, Writable, Duplex, and Transform streams.
- Streams improve performance when handling large files.

13. Advantages of Node.js

- Fast and scalable.
- Efficient for real-time applications.
- JavaScript can be used on both frontend and backend.
- Large ecosystem through npm.
- Excellent for APIs and microservices.

Exam Preparation Tips

- Prepare definitions and advantages of Node.js.
- Understand the Event Loop and asynchronous programming.
- Practice important differences and interview questions.
- Learn Express.js, Middleware, REST APIs, and JWT thoroughly.
- Revise built-in modules such as fs, http, and path.